

Satya MARTHI

*PhD Student in AI-Driven
Control Systems*

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About

PhD candidate in Control Systems and Reinforcement Learning with an aerospace engineering background, specializing in safe optimal control, reachability analysis, and nonlinear control of safety-critical systems. Experienced in MATLAB/Simulink, Python, flight-control-oriented modelling, and simulation-based validation. Research focuses on Lyapunov-based guarantees, safety filtering, and adaptive/disturbance-aware control for robotics and aerospace applications.

Education

2024– Present: **PhD, Control Systems Engineering**, *Universite de Lorraine, CRAN (Centre de recherche en automatique de Nancy)*, Nancy, France.

Funded by: Thesis funded by ANR (Agence Nationale de la Recherche) under the SOS (Self-Organizing, Smart) and safe heterogeneous robots fleet by collective emergence for a forest fire-fighting mission.

2022–2024: **Master of Science, Aerospace Engineering**, *ISAE-SUPAERO (Institut supérieur de l'aéronautique et de l'espace)*, Toulouse, France.

Major: Aerospace systems and controls.

2017–2021: **Bachelor of Technology, Aerospace Engineering**, *Amrita Vishwa Vidyapeetham*, Coimbatore, India.

Course Work: Fundamentals of Aeroelasticity, control theory, Aircraft structures, Aircraft Propulsions, Aerodynamics.

Skills

Control	Nonlinear control, optimal control, safe reinforcement learning, Lyapunov methods, control barrier functions, Hamilton-Jacobi reachability, adaptive control
Aerospace	Flight dynamics, flight control laws, handling qualities, simulation-based validation, aircraft modelling
Software	MATLAB, Simulink, Python, PyTorch, ROS, Gazebo
Tools	Reinforcement Learning Toolbox, Optimization Toolbox, GPML, Git, LaTeX

Publications

Conference Papers

- 2025 **Marthi S, Jha MS, Kanso S, Ponsart JC, Theilliol D.**, *On-policy Safe Reinforcement Learning under Input Saturation and State Constraints for Nonlinear Discrete-Time Systems*, In Proceedings of the 2025 IEEE 64th **Conference on Decision and Control (CDC)**, pp. 4817–4824, IEEE.

- 2026 **Marthi S, Jha MS, Kanso S, Theilliol D.**, *Robust Reachability within a Deep Reinforcement Learning Framework*, Accepted for presentation at the **2026 IFAC World Congress** with a journal publication option.
- 2026 **Jha MS, Vamvoudakis KG, Marthi S, Theilliol D.**, *Lipschitz-Aware Exploration for Safety in Reinforcement Learning*, In Proceedings of the 2026 IEEE **American Control Conference (ACC)**, IEEE.

Journal Papers

- 2026 **Marthi S, Jha MS, Theilliol D, Ponsart JC.**, *Robust Reachability within Deep Reinforcement Learning Framework*, Accepted for publication in **Engineering Applications of Artificial Intelligence (EAAI)**.
- 2026 **Jha MS, Marthi S, Vamvoudakis KG, Kanso S, Theilliol D.**, *Input-to-State Safety for Reinforcement Learning*, Accepted as a Regular Paper in **IEEE Transactions on Neural Networks and Learning Systems (TNNLS)**.

Journal Papers - under review

- 2026 **Marthi S, Jha MS, Theilliol D, Ponsart JC.**, *Reinforcement Learning-Based Practical Prescribed-Time Optimal Control for Nonlinear Control-Affine Systems*, To be submitted to **Automatica**.
- 2026 **Marthi S, Jha MS, Theilliol D, Ponsart JC.**, *Safe Practical Prescribed-Time Optimal Control with Input Saturation via Kronecker-Structured Neural Network Learning*, To be submitted to **IEEE Transactions on Automatic Control (TAC)**.
- 2026 **Marthi S, Jha MS, Beckers T, Theilliol D, Ponsart JC.**, *Gaussian Process based Safe Reinforcement Learning for Nonlinear Discrete-time Systems under Constraints*, To be submitted to **IEEE Transactions on Cybernetics**.

Experience

- 2024: **Research Internship**, CRAN (Centre de recherche en automatique de Nancy) , Nancy, France.
Developed and evaluated safe reinforcement learning and optimal control methods for nonlinear systems under constraints. Implemented control-barrier-function-based safety filtering and adaptive control strategies in MATLAB/Python for online optimal control with safety guarantees.
- 2022–2023 **Research Project**, ISAE-SUPAERO (Institut supérieur de l'aéronautique et de l'espace), Toulouse, France.
Modelisation of Pair of tethered Aircraft for developing flight control laws. Modelled a pair of tethered aircraft for flight-control-law development. Analyzed stability, handling qualities, and the influence of design parameters on the closed-loop behavior. Developed simulation models to support safe flight-control design.
- 2021–2022: **Internship**, STAR (Space Technology and Aeronautical Rocketry), India.
My role as an Intern is to design a static test pad using Fusion-360, with good ergonomics and aesthetics, which can be assembled and disassembled easily. Static stress analysis was performed to ensure safety

Languages

- French Intermediate proficiency - B1
- English Fluent - C1

Teaching Experience

- 2026–2027 **Third Year of PhD**, *48 teaching hours*, Polytech Nancy.
Deep Learning, Automatic Control, and Reinforcement Learning
- 2025–2026 **Second Year of PhD**, *36 teaching hours*, Polytech Nancy.
Deep Learning, Automatic Control, and Reinforcement Learning